

Factor Analysis of Mental Ability Test Scores – Case Study

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Objective of Analysis

- Goal: Identify latent dimensions underlying 9 test scores (x_1 – x_9)
- Data: 301 students, 2 schools (Pasteur, Grant-White)
- Method: Exploratory factor analysis (MLE, varimax rotation)

3 core abilities explain most of student performance.

These three dimensions together account for more than half of all variation in test scores.

Ability	What it Measures
Verbal	Reading, sentence completion, word meaning
Spatial	Visual perception, shapes, patterns
Processing Speed	Quick addition, counting, rapid decisions

Group differences detected

- How Schools & Students Differ. Schools have different profiles.
 - Pasteur : Higher spatial ability
 - Grant-White: Higher verbal ability
- Older students perform better.
 - 8th graders outscore 7th graders on verbal and spatial tasks.
- Small differences by gender.
 - Males slightly faster on processing speed.
 - Females slightly stronger on verbal tasks.

Thank
you

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